Experimental study on Biogas Production through Anaerobic Digestion of Wheat Straw and Urban Fallen Leaves

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Abstract: In this experiment, wheat straw and fallen leaves were used as the raw materials for solid-state anaerobic digestion (SS-AD). Straw, leaves and mixture of them were aerobic pretreated respectively. When the pretreated composting temperature raised to 313-328K, the raw materials were added to the test device and for SS-AD respectively. The comparison of the experiment results show that: the volatile solids (VS) gas production of the straw is higher than that of the fallen leaves. The fermentative materials of leaves have the most accumulation and the longest retention time of gas production. When the raw materials of straw and leaves at a ratio of 2:1 in the test of SS-AD, the biogas production will be increased.

Keywords: wheat straw; fallen leaves; pretreatment; solid-state anaerobic digestion

I. INTRODUCTION

With the development of economy in the rural areas of China, people's living standard improved constantly and the consumption of traditional fossil energy increased quickly. Due to China's large population, if the countryside has the same way of energy consumption as the city, it will cause big energy problems. The biomass resources is abundant in the rural areas of China, agricultural straw accounts for 70% above of the whole biomass. According to the relevant statistic in 2008, 71 million tons of agricultural straw was produced in the whole country [1], but the utilization ratio is lower than 30%. Most of them are incinerated directly in field, the environment is polluted and a lot of biomass resources are wasted. In the urban of China, the green coverage is demanded to be 48%. With the ascension of green coverage, the fallen leaves increased and the environment pollution problem caused by leaves becomes more and more serious. Therefore, it is important to use the biomass in rural areas and fallen leaves in urban city effectively [2]. This paper mainly discusses the energy utilization of wheat straw and fallen leaves in urban, and the regulation of gas production in the test of biological pretreatment and solid-state anaerobic digestion (SS-AD) is also discussed.

II. EXPERIMENTAL MATERIALS AND METHODS

A. Raw Materials and Composition Determination

Wheat straw and fallen leaves were collected in the suburb of Xi'an City and the campus of Xi'an University of Architecture and Technology respectively, the inoculums sludge was collected from the return sludge pump room of the fourth effluent treatment plant in Shannxi Province. The total solids (TS) content was measured in a drying stove at the temperature of 378±5K, the volatile solids (VS) content was measured in a muffle furnace at the temperature of 823±10K. The relevant parameters of carbon, nitrogen, cellulose, hemicellulose and lignin content were referred to the bibliography [3~5]. The composition of each raw material is shown in Table.I.

TABLE I. COMPOSITION OF RAW MATERIALS

Parameter	Straw	Leaves	Inoculum sludge
Total carbon (%)	46.5	41.0	2.7
Total nitrogen (%)	0.5	1.0	0.5
Carbon to nitrogen (C/N) ratio	97:1	41:1	5.4:1
Total solids (%)	87.7	89.5	3.6
Volatile solids (%)	84.5	87.7	4.3
Cellulose (%)	30.5	11.1	—
Hemicellulose (%)	23.5	11.5	—
Lignin(%)	18.0	22.7	—

B. Experimental Equipments and Methods

1) Pretreatment device and methods: the storage box at the size of 530×370×370mm^3 was used as pretreatment device. the raw materials were shredded at 3-5cm in the box and then modifier (303-313 K warm water) was added to wet for 24 h. After wetting, the modifier was added sequentially, and the moisture of the fermentation materials reach to 70%. It can be observed in Table.I that the ratio of C/N between straw and leaves deviate required range (20-30:1) of actual anaerobic digestion. In this experiment, we adjusted the ratio of C/N by adding the urea. Meanwhile, the raw materials were pretreated by adding bacterial agent (BM cellulose utilization microorganisms) with the ratio of 1/200. According to the temperature rise of windrow (the

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temperature rises to 313-328K), the pretreatment time was 2-4 days in summer and 7 days in winter. When the temperature of raw materials rose to 313-328K, the pretreatment stage was completed, the pretreated raw materials were placed into the anaerobic fermentation device to stand-by.

2) SS-AD device and methods: The test of device is shown in Fig.1, two jars with the volume of 2500mL (2000mL effective volume) were adopted as the fermentation device and the drainage set gas device respectively, the measuring cylinder with the volume of 1000mL was used as the catchment metering device, these three containers composed the gas connected test device. The rubber tube and the glass tube, rubber stoppers and the bottle junction of the experiment device were sealed with paraffin. The gas was collected according to the law of drainage set gas. Before the SS-AD test , the air tightness of the device should be checked. In the SS-AD test, we added 30-40% inoculated sludge to the pretreatment raw materials and regulated the TS content reach 15-30%, then placed it into device for the SS-AD test. Comparing the VS gas production of raw materials (wheat straw, leaves) at different ambient temperature, the proper SS-AD condition was obtained. The test period is from July to December in 2011, the change of water in the cylinder was measured and the rate of gas production was calculated. The gas composition was analyzed using gas chromatography of Agilent 6890N.

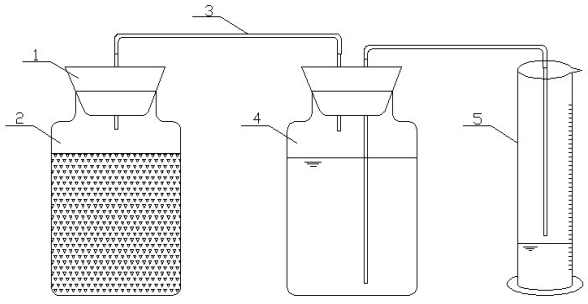


Figure 1. SS-AD test device

1-rubber stopper, 2-fermentation device, 3-rubber catheter 4-air collector 5-measuring cylinder

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Results and Analysis in Pretreatment Stage

- Change of temperature in the windrow during the bio-pretreatment process

With the process of bio-pretreatment, the temperature of windrow is rising, but after a certain reaction time, the temperature decreased gradually. The change of temperature inside the windrow was shown in Fig.2. Because the raw materials was pre-wetted before the pretreatment, the temperature in the windrow raised quickly. After pretreated for 4 days, the windrow reached the highest temperature of 326K. The temperature in the windrow was above 313K for7 days before pretreating. After pretreated for 8 days, with the SS-AD test loading process, the stir intensity strengthened, the heat released from the windrow increased gradually, the temperature in the windrow dropped quickly until the same as the indoor temperature. In this experiment, the raw materials of the SS-AD test were selected at the pretreatment time of 4 days and 7 days respectively.

- Influence of gas production on the bio-pretreatment time and the raw materials

The pretreatment time was different, the cumulative gas production in the SS-AD process was also different. The pretreatment time for 4 days (feed temperature was 326K) and 8 days (feed temperature was 301K), the cumulative gas production was shown in Fig.3. The lignin and the waxy layer of the wheat straw were not destroyed completely after the shorter pretreatment time of 4 days, but at this time the windrow reached the highest temperature of 326K. With the temperature of 326K windrow to loading for the SS-AD, the anaerobic digestion with high start-up temperature, the thermophilic and anaerobic bacterium made the raw material degradation rapidly, the raw materials can produce gas normally. The gas production was 280mL initially, the duration of gas production was 28 days, when the cumulative gas production reached 6400mL, the gas production finished. after pretreated for 8 days, the temperature in the windrow declined to 301 K, loading the raw material. The pretreatment time is longer, the lignin and waxy layer is further damaged and broken down, but the organic matter in the straw also degraded rapidly. At the beginning of SS-AD, the initial gas production was 460mL, after the cumulative gas production reached to 3332mL, the gas production duration was 13 days, the raw materials of fermentation stop producing gas due to the lack of organic matter. Therefore, when pretreated for 4 days and the raw materials temperature reached to 326K, loading the raw materials can obtain the most cumulative gas production and the longest gas production time. It implied that the reasonable pretreatment time and the temperature of raw materials have great influence on gas production. Increasing the raw materials temperature of the SS-AD appropriately will increase the biogas yield.

B. Results and Analysis about the Process of SS-AD

- Influence of ambient temperature on the biogas production

In this SS-AD experiment, the reaction equipment was placed in room, the ambient temperature would affect the gas production of SS-AD directly. The change of biogas production with ambient temperature as shown in Fig.4. When the ambient temperature rises, the gas production of SS-AD process also increased gradually. When the ambient temperature reached to 305.5K, the biogas quantity was 710mL, when the ambient temperature dropped to 292K, the biogas quantity was only 30ml. When the ambient temperature changed moderately, the change range of biogas production was not too much. When the ambient temperature changed above 278K, the gas production changed obviously, and sometimes even stop producing gas. However, when the ambient temperature raised again, the gas production rebounded. It can be seen that the change of ambient

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temperature has great influence on the SS-AD process, it will increase the biogas production effectively if you raise the ambient temperature appropriately.

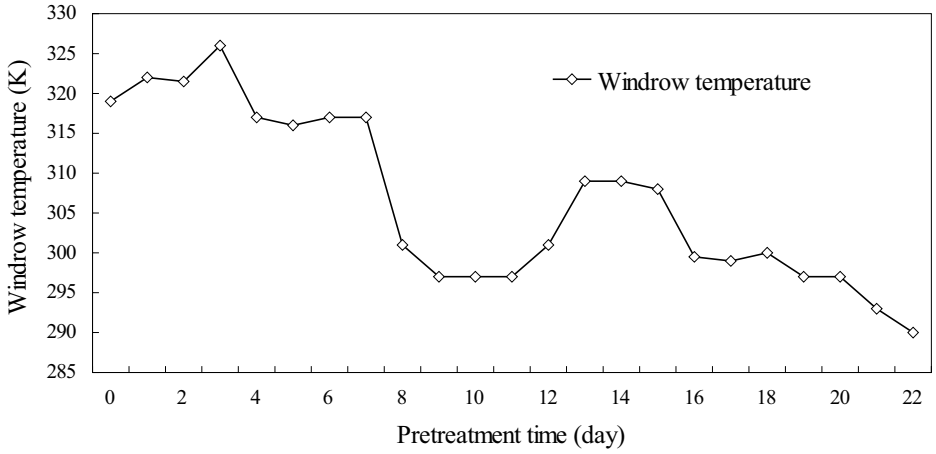


Figure 2. The change of windrow temperature during the bio-pretreatment process

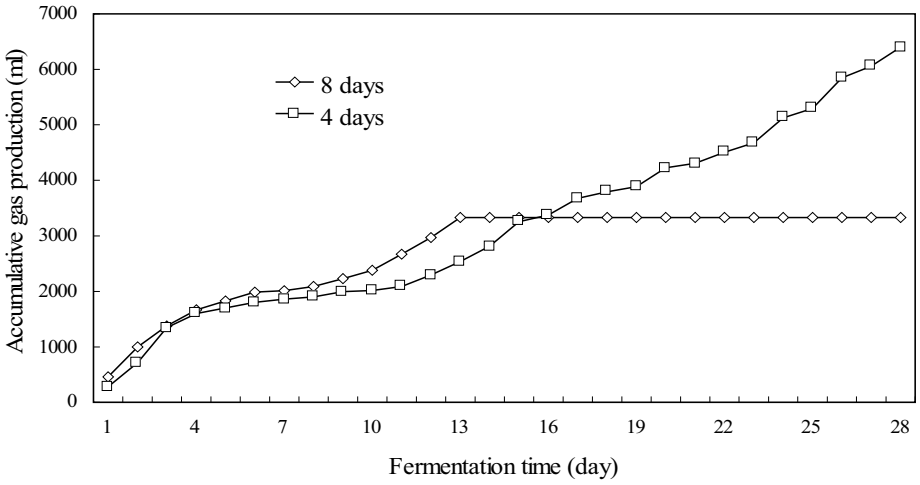


Figure 3. The change of cumulative biogas production at different pretreatment time

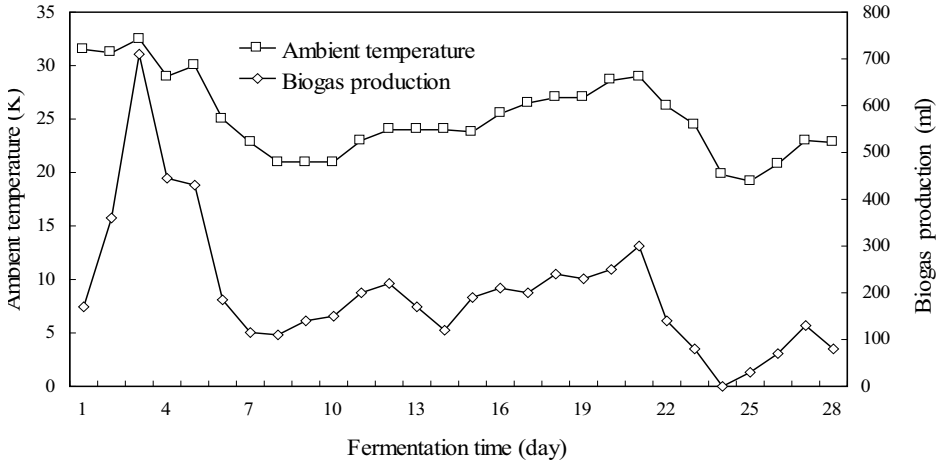


Figure 4. The change of biogas production at different ambient temperature

- Influence of different raw materials on biogas yield

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The gas yield of SS-AD process is different when using different biomass materials. Selecting appropriate materials is the most important for biogas yield. In this experiment, we selected three materials, straw, leaves and mixture of them (mixed at a ratio of 2:1) for contrast. The cumulative biogas yield of each material in the SS-AD process was shown in Fig.5. The parameters of each material were shown in Table II. The methane content of biogas was above 50% when the reaction at the third day with different ratio of raw materials for SS-AD.

TABLE II. BIOGAS YIELD PARAMETERS OF EACH MATERIAL AFTER SS-AD PROCESS

Parameter	Straw	Leaves	Straw: leaves=2:1
Accumulative gas production (mL)	4460	9375	6416
VS gas production (L/kg•VS)	165.19	97.66	92.99
Retention time of gas production (d)	21	52	31
Gas production per unit volume (L/L)	2.23	4.69	3.21
Daily average gas production (L/kg•d)	1.93	0.86	0.92

As shown in Fig.5 and Table II, the gas production of wheat was the highest, so is the average daily gas production. The accumulative gas production leaves was the highest and the gas production duration is the longest. Leaves and wheat straw co-digestion of raw materials accumulation gas production more than mono-digestion of leaves 1956mL, and the daily gas production is higher than 0.06L/kg•d, gas production of unit volume was higher than mono-digestion wheat straw 0.98L/L.

The reason is that cellulose and hemicellulose in the leaves (11.1%, 11.5%) is lower than wheat straw (30.5%, 23.5%), while the lignin content of leaves is 22.7% and that of the wheat straw is 18.0% (see Table.1). The studies of Lo Niee Liew [6] have shown that the biogas production ability of raw materials is directly influenced by the degradation extent of cellulose and hemicellulose. For the raw materials, the content of cellulose and hemicellulose is higher, the degradation ability of raw materials is stronger during the pretreatment process, the gas production will increase. The VS gas production of wheat straw with higher content of cellulose and hemicellulose is higher than that of leaves.

However, after shredded the wheat straw and leaves, leaves have higher packing density than wheat straw, when loading the materials, the per unit volume of leaves was 0.91kg more than 0.53kg of wheat straw, and the leaves have the most accumulative gas production. Besides, the studies of Deepak Singh [7] have shown that the lignin in the plant cell wall connection with cellulose and hemicellulose to provide plant has integrate structural. It can account for higher content of lignin, the slower degradation rate of raw materials. In the pretreatment process broken the lignin in the cell wall can facilitate degradation of cellulose and hemicellulose, increase the gas production. Because of the higher lignin content in the leaves, the degradation rate in pretreatment time is lower, making the leaves have longer time of gas production in SS-AD process and more accumulative gas production. Co-digestion of leaves and wheat straw, because of the particular structure in the leaves, the per unit volume gas production of raw materials was higher than mono-digestion wheat straw, daily gas production was higher than mono-digestion of leaves. Therefore, according to the actual situation, adding the leaves as the raw materials in the SS-AD process, will not only extend the SS-AD time, but also increase the biogas production.

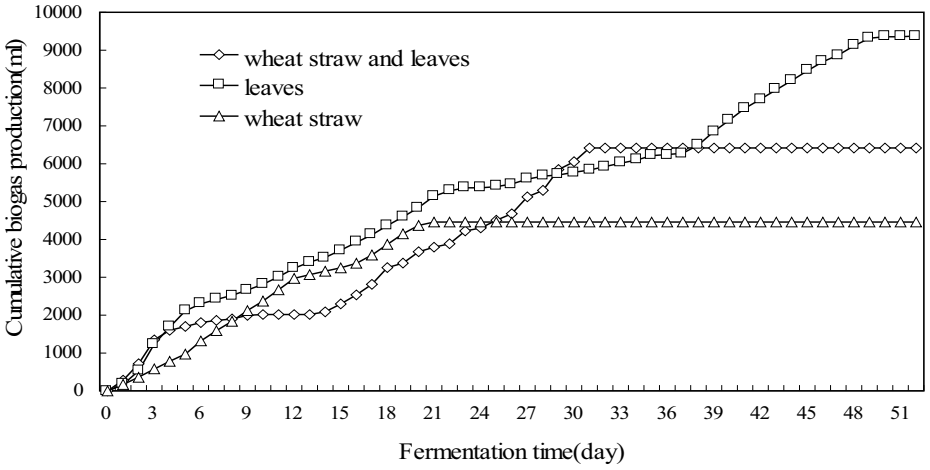


Figure 5. The change of accumulative biogas production for different raw materials

IV. CONCLUSIONS

1) The experiment results show that pretreating the raw materials with physical grinding and biological agent, can effectively broke down the lignin and waxy layer of biomass straw . Besides, it can also increase the contact area of cellulose and hemicellulose, and increase the degradation of organic matter to improve biogas production. It can increase the biogas production effectively when dry fermentation directly with the pretreatment temperature at 326K. It indicates that the reasonable pretreatment temperature has great influence on accumulative gas production.

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2) Ambient temperature is the important factor for the SS-AD process. When the ambient temperature reaches above 303 K, the biogas production will improve effectively. For the actual operation of the SS-AD process, increasing the ambient temperature can increase the accumulative biogas production.

3) Compared with wheat straw and fallen leaves, as the cellulose and hemicellulose of wheat straw is higher than that of leaves, the mono-digestion of straw's VS gas production is higher than that of leaves, which is 67.53L/kg·VS. However, due to the packing density in the leaves is higher and per unit volume loading the raw materials is larger, the leaves have the most accumulative gas production and the longest production time.

4) In the wheat straw's SS-AD process, adding leaves as the raw materials appropriately not only can increase the accumulative biogas production, but also extend reaction time of SS-AD process. Meanwhile, it can effectively use urban leaves and solve the environmental problems in urban areas.

ACKNOWLEDGMENT

This work was supported by the "13115" Key Project of Technology Innovation for the New Rural Reconstruction of Shaanxi Province through No.2009ZDKG-66.

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